

# Observer-Patch Holography and the Dark Matter Phenomenon

## A Repair-Charge Condensate Completion

Bernhard Mueller

David Matscheko

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### Abstract

Observer-patch holography contains a distinguished scalar repair register but does not assign that register a propagation law. This paper states a candidate dynamical completion: integer repair occupation and a conjugate repair phase form a lattice rotor. The proposed action has an exact repair-current balance law and a metric stress tensor. Its dilute homogeneous phase is pressureless, while a cubic condensed-phase link energy gives the deep radial-acceleration law and the baryonic Tully–Fisher scaling. Coherent material couples through the canonical susceptibility  $\chi_\nu^{\text{can}}$ , producing a lawful force only through genuine repair charge, an external repair gradient, or explicit field-momentum exchange. The canonical pair, source normalizations, complete constitutive law, relativistic limit, abundance, lensing, CMB, cluster, and laboratory receipts are work in progress. The construction is a conditional dark-sector mechanism, not a closed cosmological prediction.

## Claim Boundary

The finite OPH results supply the scalar register, the coherent-material same-channel theorem, and the protected-reserve coefficient

$$\chi_\nu^{\text{can}} = 1 - \frac{P}{24} = 0.9320429912748350 \dots \quad (1)$$

They do not supply a canonical conjugate, kinetic term, stress tensor, baryonic source map, or dimensional coherent-source coupling.

The repair-charge condensate action below is a proposed completion. Given that action, its continuity equation, stress balance, dilute-fluid equation of state, spherical deep-gradient law, and device-force ledger are mathematical consequences. Deriving the action from finite OPH repair dynamics is work in progress. Cosmology remains outside the OPH falsification program.

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## 1 The Missing Dynamical Object

A scalar channel identifies where a perturbation lives. Dynamics requires a canonical pair. Let the fixed-cutoff OPH carrier be an oriented causal cell complex with sites  $v$ , oriented links  $e$ , metric-dependent weights  $V_v$  and  $W_e$ , integer signed repair occupation  $n_v$ , and conjugate phase  $\theta_v$ . The pair obeys

$$n_v \in \mathbb{Z}, \quad \theta_v \in S^1. \quad (2)$$

The local formulae use a real lift of  $\theta$ . A global compact-phase completion must make the exponentiated source coupling single-valued. That quantization condition is work in progress.

Let  $S_{\text{coh},v}^{\text{can}}$  be the canonical coherent-material source and  $\mathcal{Q}_{b,v}$  the baryonic repair demand. The fixed-chart Lorentzian action is

$$\begin{aligned} I_{\text{RPC}} = I_{\text{Regge}}[\ell] + I_{\text{SM}}[\ell, \Psi] \\ + \int d\tau \left[ - \sum_v V_v n_v D_\tau \theta_v - \sum_v V_v \varepsilon(n_v) - \sum_e W_e \mathcal{K}(|(d\theta)_e|) \right. \\ \left. - \sum_v V_v \theta_v \left( \mathcal{Q}_{b,v} + q_\star \chi_\nu^{\text{can}} S_{\text{coh},v}^{\text{can}} \right) \right]. \end{aligned} \quad (3)$$

Here  $q_\star$  is the dimensional coherent-source calibration. The infrared constitutive functions are

$$\varepsilon(n) = m_R |n| + \frac{u_R}{3} |n|^3 + \cdots, \quad \mathcal{K}(y) = \frac{\kappa_R}{3} y^3 + \cdots, \quad (4)$$

with  $m_R, u_R, \kappa_R > 0$ . The omitted terms must provide a positive convex high-gradient completion.

Let  $D_\tau n_v := V_v^{-1} \partial_\tau (V_v n_v)$ , and define the lattice divergence by

$$\sum_v V_v f_v (\text{div}_R j)_v = - \sum_e W_e (df)_e j_e \quad (5)$$

for compactly supported  $f$ . These conventions make phase variation produce a positive source, and translation of a positive localized source gives  $-Q_R \nabla \theta$ .

## 2 Repair Charge, Current, and Stress

Variation with respect to the phase gives

$$D_\tau n + \text{div}_R j_R = \mathcal{Q}_b + q_\star \chi_\nu^{\text{can}} S_{\text{coh}}^{\text{can}}, \quad (6)$$

where

$$j_{R,e} = \mathcal{K}'(|(d\theta)_e|) \frac{(d\theta)_e}{|(d\theta)_e|}. \quad (7)$$

For the cubic infrared branch,

$$j_{R,e} = \kappa_R |(d\theta)_e| (d\theta)_e. \quad (8)$$

Summation over a finite region  $\Omega$  gives the exact lattice balance law

$$\frac{d}{d\tau} \sum_{v \in \Omega} V_v n_v + \sum_{e \in \partial\Omega} W_e j_{R,e} = \sum_{v \in \Omega} V_v \left( \mathcal{Q}_{b,v} + q_\star \chi_\nu^{\text{can}} S_{\text{coh},v}^{\text{can}} \right). \quad (9)$$

Equation (9) leaves three lawful source classes. A compact neutral source has no monopole and begins at dipole order. A source with nonzero integrated repair charge has a noncompact field tail. A driven source exchanges repair charge and momentum through its boundary. A compact neutral scalar contrast cannot produce a static support force by itself.

In the static cubic branch the spatial stress is

$$T_{ij}^R = \kappa_R |\nabla\theta| \partial_i \theta \partial_j \theta - \delta_{ij} \frac{\kappa_R}{3} |\nabla\theta|^3 + \dots. \quad (10)$$

Translation invariance gives the momentum ledger

$$F_i^{\text{matter}} + \frac{dP_i^R}{dt} + \oint_{\partial\Omega} T_{ij}^R n^j dA = 0. \quad (11)$$

## 3 Dilute Homogeneous Phase

Variation with respect to  $n_v$  gives

$$D_\tau \theta_v = -\varepsilon'(n_v). \quad (12)$$

On the positive homogeneous branch,

$$\rho_R = m_R n + \frac{u_R}{3} n^3, \quad p_R = n \varepsilon'(n) - \varepsilon(n) = \frac{2u_R}{3} n^3. \quad (13)$$

For  $u_R n^2 \ll m_R$ , the repair medium satisfies

$$p_R \ll \rho_R, \quad \rho_R \simeq m_R n. \quad (14)$$

With no active source, Eq. (6) gives repair-charge conservation. In a homogeneous expanding chart,

$$\dot{n} + 3Hn = 0, \quad n \propto a^{-3}, \quad \rho_R \propto a^{-3}, \quad w_R \simeq 0. \quad (15)$$

The proposed repair medium therefore contains a cold-dark-matter-like background phase. Its abundance and perturbation transfer functions are not fixed by Eq. (15).

## 4 Condensed Galactic Phase

In a static condensed region, take

$$\nabla \cdot (\kappa_R |\nabla \theta| \nabla \theta) = \mathcal{Q}_b, \quad \mathcal{Q}_b = \beta_b \rho_b, \quad \mathbf{a}_R = -\beta_b \nabla \theta. \quad (16)$$

For a spherical baryonic source of mass  $M_b$ , the exterior attractive branch satisfies

$$4\pi r^2 \kappa_R |\theta'| \theta' = \beta_b M_b, \quad (17)$$

and hence

$$a_R = \sqrt{\frac{\beta_b^3 M_b}{4\pi \kappa_R}} \frac{1}{r}. \quad (18)$$

Define

$$a_0 = \frac{\beta_b^3}{4\pi G \kappa_R}. \quad (19)$$

Then

$$\boxed{a_R = \frac{\sqrt{GM_b a_0}}{r} = \sqrt{a_b a_0}}, \quad \boxed{v^4 = GM_b a_0}. \quad (20)$$

Thus the cubic link energy implies the deep radial-acceleration relation and the baryonic Tully–Fisher scaling on the stated spherical, static, condensed-phase premises. The complete interpolation, nonspherical dynamics, external-field behavior, and value of  $a_0$  require the full microscopic function  $\mathcal{K}$  and the source-derived constants  $\beta_b, \kappa_R$ .

A single medium with a pressureless normal phase and a MOND-like condensed response also appears in superfluid dark-matter models [4]. Dipolar media provide another action-based example in which cosmological dust behavior and galactic polarization coexist [5]. These analogies establish mathematical precedent, not an OPH derivation.

## 5 High-Gradient Branch

The constitutive completion may have

$$\mathcal{K}(y) \sim \frac{\kappa_R}{3} y^3 \quad \text{at low gradient}, \quad \mathcal{K}(y) \sim \frac{\kappa_\infty}{p} y^p, \quad p > 3, \quad \text{at high gradient.} \quad (21)$$

For fixed source flux, the high-power branch makes the field gradient grow sublinearly with source strength. This is a kinetic-screening route rather than a system label. K-mouflage models show that nonlinear kinetic screening can coexist with Solar-System and cosmological constraints, subject to strong parameter bounds [6]. Equation (21) does not establish Solar-System viability. The Sun-plus-Galaxy boundary problem and the relativistic coupling are work in progress.

## 6 Coherent Matter and $\chi_\nu$

The coherent source in Eq. (3) is

$$\mathcal{Q}_{\text{coh}} = q_\star \chi_\nu^{\text{can}} S_{\text{coh}}^{\text{can}}. \quad (22)$$

The coefficient  $\chi_\nu^{\text{can}}$  is the dimensionless efficiency with which the canonical coherent source enters the repair-charge channel. It is not a complete force coefficient. The dimensional normalization  $q_\star$ , the ambient repair field, and the source charge are separate objects.

If an engineering chart uses

$$S_{\text{coh}}^{\text{eng}} = N_{\text{coh}} S_{\text{coh}}^{\text{can}}, \quad (23)$$

then

$$\chi_\nu^{\text{eng}} = \frac{\chi_\nu^{\text{can}}}{N_{\text{coh}}}. \quad (24)$$

This is a coordinate change. It fixes neither  $q_\star$  nor a device force.

A coherent source in an external repair field experiences

$$F_i^{\text{coh}} = -q_\star \chi_\nu^{\text{can}} \int_{\Omega} S_{\text{coh}}^{\text{can}}(x) \partial_i \theta_{\text{ext}}(x) \text{d}^3 x. \quad (25)$$

Define

$$Q_R^{\text{dev}} = q_\star \chi_\nu^{\text{can}} \int_{\Omega} S_{\text{coh}}^{\text{can}} \text{d}^3 x. \quad (26)$$

For a slowly varying external field,

$$F_i^{\text{dev}} \simeq -Q_R^{\text{dev}} \partial_i \theta_{\text{ext}}. \quad (27)$$

Signed repair charge permits attractive or repulsive response while  $\varepsilon(n) = \varepsilon(|n|)$  keeps the onsite energy positive.

A local repulsive response therefore requires either

$$Q_R^{\text{dev}} \neq 0 \quad (28)$$

with its field tail, or nonzero field-momentum transfer in Eq. (11). A static, closed, repair-neutral device cannot support itself through an internal top-to-bottom coherence contrast.

## 7 Physical Completion Conditions

The conditional mechanism becomes an OPH dark-sector theory only if one construction supplies all of the following:

1. a compact phase canonically conjugate to the scalar repair occupation;
2. a globally well-defined source coupling for the compact phase;
3. the baryonic source map  $\mathcal{Q}_b$  from the modular/Borel–Weil gravity bridge;
4. the dimensional coherent-source calibration  $q_\star$ ;
5. a positive convex constitutive law  $\mathcal{K}$  covering galactic and high-gradient regimes;
6. a relativistic refinement limit and complete conserved stress tensor;
7. a source-derived cosmic repair abundance and perturbation initial data;

8. CMB, lensing, cluster, merger, growth, and Solar-System likelihoods;
9. an instrument-independent coherent-source observable;
10. a laboratory receipt for nonzero repair monopole charge or outgoing repair-field momentum.

No cosmological target or laboratory lift value is registered. Diagnostic simulations and known-data comparisons carry no falsification verdict while these constructions are work in progress.

## 8 Conclusion

The proposed scalar repair register is a dynamical rotor with occupation  $n$  and phase  $\theta$ . Its normal phase has pressureless background scaling. Its cubic condensed phase gives the deep galaxy law. Coherent matter enters through the same scalar channel with coefficient  $\chi_\nu^{\text{can}}$ , while a lawful device force requires repair charge, an external gradient, and a closed momentum ledger. This unifies the dark-sector and coherent-material mechanisms at the level of one conditional action. The OPH derivation and physical evidence are work in progress.

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